INCH-POUND

MIL-M-44074E
31 December 1992
SUPERSEDING
MIL-M-44074D
17 December 1990

MILITARY SPECIFICATION

MEAL, READY-TO-EAT, INDIVIDUAL, ASSEMBLY OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the assembly requirements for the Meal, Ready-To-Eat, Individual.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

FEDERAL

A-A-461 - Hand Cleaner (Premoistened Paper Towelette in Packet)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8970

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A-A-20041	-	Salt, Table, Iodized
A-A-20097	-	Hot Sauce
A-A-20098	-	Beverage Bases (Powdered)
A-A-20112	-	Noodles, Chow Mein
A-A-20135	-	Sugar, Refined and Sugar, Brown
A-A-20175	-	Chewing Gum
A-A-20177	-	Candy and Chocolate Confections
A-A-20195	-	Potato Sticks
L-F-560	-	Fork, Knife, and Spoon, Picnic, Plastic
L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
EE-M-101	-	Matches, Safety: and Matches, Nonsafety (Strike-Anywhere)
QQ-A-1876	-	Aluminum Foil
HHH-C-575	-	Coffee, Instant
MMM-A-250	-	Adhesive, Water-Resistant (for Closure of Fiberboard Boxes)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-C-3031	-	Cocoa Beverage Powder
MIL-B-35023	-	Beverage Base, Powdered - (Operational Ration Component)
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-43338	-	Cream Substitute, Dry, Non-Dairy
MIL-P-43988	-	Paper, Toilet, Tissue (Packet in Sheet Form)
MIL-B-44059	-	Beef Stew, Thermostabilized, For Meal, Ready-To-Eat
MIL-F-44062	-	Frankfurters, Beef, Thermostabilized, For Meal, Ready-To-Eat
MIL-H-44063	-	Ham Slice, Smoke Flavoring Added, Thermostabilized, For Meal, Ready-To-Eat
MIL-F-44067	-	Fruits, Thermostabilized, For Meal, Ready-To-Eat
MIL-C-44068	-	Cheese Spread, Cheddar; Peanut Butter; and Jelly, For Meal, Ready-To-Eat
MIL-F-44070	-	Fruits, Freeze-Dehydrated, For Meal, Ready-To-Eat
MIL-C-44071	-	Cakes, Thermostabilized, For Meal, Ready-To-Eat
MIL-C-44072	-	Cookies, Oatmeal; and Brownies, Chocolate Covered, For Meal, Ready-To-Eat
MIL-C-44112	-	Crackers, For Meal, Ready-To-Eat
MIL-O-44136	-	Oatmeal Cookie and Granola Bars - (Operational Ration Component)
MIL-E-44200	-	Escalloped Potatoes with Ham, Thermostabilized, For Meal, Ready-To-Eat
MIL-P-44201	-	Pork with Rice in Barbecue Sauce - Smoke Flavoring Added, Thermostabilized, For Meal, Ready-To-Eat
MIL-O-44202	-	Omelet with Ham, Thermostabilized, For Meal, Ready-To-Eat

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MIL-C-44204 - Corned Beef Hash, Thermostabilized, For Meal, Ready-To-Eat
MIL-S-44205 - Spaghetti with Meat and Sauce, Thermostabilized, For Meal, Ready-To-Eat
MIL-P-44206 - Potatoes Au Gratin, Thermostabilized, For Meal, Ready-To-Eat
MIL-T-44208 - Tuna with Noodles, Thermostabilized, For Meal, Ready-To-Eat
MIL-C-44209 - Chicken Stew, Thermostabilized, For Meal, Ready-To-Eat
MIL-C-44210 - Chicken and Rice, Thermostabilized, For Meal, Ready-To-Eat
MIL-R-44398 - Ration Supplement, Flameless Heater, For Meal, Ready-To-Eat
MIL-P-44432 - Pork Chow Mein, Thermostabilized, For Meal, Ready-To-Eat

STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

13-1-0183 - Figure 1. Accepted Meal Bag Packing Configurations

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: SATNC-UX, Natick, MA 01760-5018.)

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U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR PARTS 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1238 - Flow Rates of Thermoplastics by Extrusion Plastometer
D 1505 - Density of Plastics by Density Gradient Technique
F 88 - Seal Strength of Flexible Barrier Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Components.

3.1.1 Meal bag. The meal bag shall be made from polyethylene tubing that is approved for food use and is average gauge 0.010 to 0.012-inch thick, with a density of 0.910 to 0.929 g/cm^3. No area of the bag, including gel-induced depressions, shall be less than 0.010-inch thick. The color of the polyethylene shall be in the range of numbers 34079 through 34102, or 24052 through 24087, or 30045 through 30118 (excluding 30109) or shall conform to number 10045, 20040, 20045, 20062, or 34052 of FED-STD-595. The inside width of the bag shall be not less than 7-1/4 inches and not more than 7-3/4 inches and the inside length shall be not less than 11-3/4 inches and not more than 12-1/8 inches. The bag opening shall be across the width. The

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bag manufacturer's seal shall be a minimum 1/16 inch wide and located with the outside edge of the seal 1/2 inch (± 1/16 inch) from one end of the bag. The seal shall form a hermetic closure. The average seal strength of each seal shall be not less than 9 pounds per inch of width and no individual specimen shall be less than 7 pounds (see 4.2.1.2). A 1/4 to 3/8 inch deep slit or V-shaped tear notch shall be located at one end of the bag within 1/2 inch of the longitudinal center line of the bag.

3.1.1.1 Meal bag marking. Each meal bag shall be printed on at least one face in permanent ink in clear black letters not less than 3/16 inch high, as follows:

MEAL, READY-TO-EAT, INDIVIDUAL

MENU NO. (1 through 12, as applicable)

_____

(Name of applicable meat component as listed in table I component column)

(name and address of assembler)

Continuous printing is acceptable provided that the complete markings appear uninterrupted on at least one face of the bag.

3.1.2 Accessory packet. The accessory packet shall be fabricated in accordance with 3.1.2.1 or 3.1.2.2.

3.1.2.1 Preformed packets. The preformed accessory packet shall be fabricated from material made by hot-extruding or laminating 0.0015-inch thick (± 10 percent) polyethylene on 0.00035-inch thick (± 10 percent) aluminum foil and laminating the opposite side of the foil to 0.0005-inch thick (± 20 percent) polyester. The color of the exterior (polyester) side of the laminate shall conform to number 30045, 30099, 30108, 30111, 30117, 30118, 34052, 24064, 34079, 34086, or 34087 of FED-STD-595. Any difference in gloss caused by the polyester exterior layer will be considered acceptable. The inside width of the packet shall be not less than 3-5/8 inches and not more than 5-1/8 inches. The inside length of the packet shall be not less than 4-3/8 inches and not more than 7-3/8 inches. The dimensions selected from the allowable ranges shall be sufficient to contain all the components for the particular accessory packet (A or B) being produced, and to effect compatibility with the equipment used for packet fabricating, filling, and sealing equipment. Each packet shall be heat sealed on three edges with 3/8 inch wide (± 1/8 inch) seals. A 1/8 inch wide (± 1/16 inch) lip may be incorporated at the open end of the packet to facilitate opening and filling of the packet. A tear nick or notch shall be located on the outer edge of one or both side seals to facilitate easy opening of the

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packet. Each tear nick or notch shall be a minimum 1/16 inch deep, but shall not extend into the side seal to a depth that reduces the seal width to less than 1/8 inch. The average seal strength of the packet seals shall be not less than 3.5 pounds per inch of width and no individual specimen shall have a seal strength of less than 3.0 pounds per inch of width (see 4.2.1.2).

3.1.2.2 Form-fill-seal packet. The form-fill-seal accessory packet shall consist of a tray-shaped body with a tray-shaped or flat sheet heat sealable cover. The tray-shaped body and tray-shaped cover (when used) shall be formed by drawing flexible laminated material into an appropriately shaped cavity. The tray-shaped or flat sheet cover shall be applied by a continuous hermetic heat seal along the entire perimeter of the tray-shaped body. The tray-shaped body and tray-shaped cover (when used) of each packet shall be fabricated from 0.0015-inch thick linear low density polyethylene bonded to 0.0005-inch thick aluminum foil with 7 pound per ream low density polyethylene extrusion resin and 0.00075-inch thick oriented polypropylene bonded to the opposite side of the aluminum foil with 10 pounds per ream low density polyethylene extrusion resin. The flat sheet cover (when used) of each packet shall be fabricated from 0.0015-inch thick linear low density polyethylene bonded to 0.00035-inch thick aluminum foil with 7 pound per ream low density polyethylene extrusion resin and 0.0005-inch thick polyester bonded to the opposite side of the aluminum foil with 10 pound per ream low density polyethylene extrusion resin. The linear low density polyethylene shall be the copolymer of ethylene and octene-1, having melt index range of 0.8 to 1.2 g/10 minutes in accordance with ASTM D 1238 and a density range of 0.918 to 0.922 g/cc in accordance with ASTM D 1505. The color of the exterior side of the laminates, (polypropylene side of the tray-shaped body and tray-shaped cover and the polyester side of the flat sheet cover) shall be as specified in 3.1.2.1. The inside width and length dimensions of the packet shall be as specified in 3.1.2.1. The closure seal at the perimeter of the packet body and cover shall be a minimum 1/4 inch wide. A tear nick or notch shall be located on one or both outside edges of the packet length or width to effect easy opening of the packet in the machine direction of the packet body and cover laminates. Each tear nick or notch shall be a minimum 1/16 inch deep, but shall not extend into the seal to a depth that reduces the seal width to less than 1/8 inch. Alternatively, if the packet has serrated edges, the serrations may be used as tear notches provided that the serrations are located to effect easy opening in the machine direction of the packet laminates; the serrations are sharp (no plastic tailings exist); and the serration depth and minimum seal width at the serrations are as specified for the tear nick or notch. The average seal strength of the packet seals shall be not less than 3.5 pounds per inch of width, and no individual specimen shall have a seal strength of less than 3.0 pounds per inch of width when tested in accordance with 4.2.2.2. Alternatively, the form-fill-seal accessory packet may be made on equipment that does not change the shape of the bag laminate material from the flat sheet form. The body shall not be formed into a tray, it shall be in the form of a flat sheet like the cover. When this equipment is employed, the packet shall be made in accordance with the requirements of 3.1.2.1, except that all four sides of the packet bag

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shall be sealed in the sequence required by the equipment used. Projection of the packet bag material beyond the outside edges of the seals is permissible. Alternatively, the form-fill-seal accessory packet may be constructed as a three seamed pillow pouch, utilizing a back seam which joins the opposite edges of the packaging materials into a tube. The back seam shall be a fin seal which is formed by heat sealing the two opposite inside edges to each other and folding the fin seal flat, in the plane of the material. The tube may be formed around the contents being packed or it may be formed empty and filled in a subsequent step. The material specified in 3.1.2.1 shall be used. Tear nicks, notches, or serrations shall be provided on one or both end seals.

3.1.2.3 Accessory packet marking. The letter of each packet shall be printed on one or both sides in permanent, black ink:

(A or B as specified)

3.1.3 Menu components. The components required for menu assembly shall be as listed in table I.

TABLE I. Menu components

Component	Document reference
Beef stew	MIL-B-44059
Frankfurters, beef	MIL-F-44062
Chicken stew	MIL-C-44209
Chicken and rice	MIL-C-44210
Corned beef hash	MIL-C-44204
Escalloped potatoes with ham	MIL-E-44200
Omelet with ham	MIL-O-44202
Ham slice	MIL-H-44063
Pork chow mein	MIL-P-44432
Pork with rice in barbecue sauce	MIL-P-44201
Spaghetti with meat and sauce	MIL-S-44205
Tuna with noodles	MIL-T-44208
Potatoes au gratin	MIL-P-44206
Applesauce	MIL-F-44067
Fruits, wet pack	MIL-F-44067 1/
Fruits, freeze-dehydrated 2/	MIL-F-44070
Pound cake	MIL-C-44071 3/
Chow mein noodles	A-A-20112
Potato sticks	A-A-20195
Cheese spread	MIL-C-44068
Jelly (apple, grape) 4/	MIL-C-44068
Peanut butter	MIL-C-44068
Brownie, chocolate covered 5/	MIL-C-44072
Cookie, chocolate covered 5/	MIL-C-44072

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TABLE I. Menu components (cont'd)

Component	Document reference
Oatmeal cookie bar	MIL-O-44136, type I
Cocoa beverage powder	MIL-C-3031, type I, class I
Crackers 6/	MIL-C-44112
Accessory packet, A or B	3.1.2 and 3.1.4
Beverage base, powdered 4/	MIL-B-35023, flavors 1 through 6
Hot sauce	A-A-20097, type II, 1/8 fluid ounce bottle
Chocolate disks, sugar, pan coated 7/	A-A-20177, type VI
Hard candy, fruit tablets 7/	A-A-20177, type IV
Milk chocolate bar, heat resistant 7/	A-A-20177, type VIII, 1 ounce bar
Sugar-free beverage base 8/	A-A-20098, type I, class D, flavors 7, 8 and 12, eight-ounce yield envelope
Spoon	L-F-560, type 4, item 13
Flameless ration heater (FRH)	MIL-R-44398

1/ Type III, IV or V of MIL-F-44067 shall be used.

2/ At least three of the four different kinds of freeze-dehydrated fruit (fruit mix, peaches, pears, strawberries) shall be distributed in each shipping container.

3/ Types VII, VIII, IX, X, and XI shall be procured in equal quantities and distributed in as uniform a distribution as possible in each shipping container. However, not more than two meal bags in any shipping container shall contain the same type of cake.

4/ To be procured in equal quantities and assembled into appropriate menus in as uniform a distribution as possible in each shipping container.

5/ When cookies or brownies are supplied to the ration assembler in bulk packs (see 6.1), one brownie or one cookie shall be unit packed in accordance with the applicable requirements of MIL-C-44072. Bulk packed brownies and cookies shall not be stored above 80°F and if storage time exceeds 30 days, they shall be stored at 0°F or below. Frozen cookies or brownies shall be tempered for approximately 24 hours at 65° ± 5°F and not more than 55 percent relative humidity prior to insertion into cookie/brownie bags.

6/ When crackers are supplied to the unit packager in bulk packs (see 6.1), two crackers shall be unit packed in accordance with the applicable requirements of MIL-C-44112. Crackers shall not be more than 90 days old from date of manufacture to date of unit pack.

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7/ Three packages of chocolate disks, two heat stable chocolate bars, and one package of fruit tablets shall be procured for each shipping container of assembled MREs.

8/ Flavors 7, 8 and 12 shall be procured in equal quantities and distributed in as uniform distribution as possible in menus 1, 2, 4, 5, 6, and 12.

3.1.4 Accessory packet components. The components required for accessory packet assembly shall be as listed in table II.

TABLE II. Accessory packet components

Component	Document reference	Used in packet
Coffee, instant	HHH-C-575, type III, style A, blend (1), composition 1	A and B
Cream substitute	MIL-C-43338	A and B
Sugar	A-A-20135, type I, class 1, 6-gram envelope	A and B
Salt	A-A-20041, iodized, 4-gram envelope	A and B
Chewing gum	A-A-20175, type I, flavors 1 and 2 1/	A and B
Matches	EE-M-101, type II, class 2	A and B
Toilet tissue	MIL-P-43988	A and B
Hand cleaner (pre-moistened paper towelette in packet)	A-A-461	A and B
Candy: Vanilla caramel pieces (or) 1/	A-A-20177 - type I	B 2/
Chocolate rolls 1/	- type II	B 2/
Hot sauce	A-A-20097, type II, 1/8 fluid ounce bottle	A and B 3/

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1/ Each shipping container of assembled MREs shall have 2 B accessory packets. Candy shall be procured so that 1 of the B accessory packets contains chocolate toffee rolls and the other B accessory packet contains vanilla caramels.

2/ In lieu of packing candies in accessory packet B, they may be unit packed separately in a heat sealable barrier material one layer of which is a minimum of 0.00025-inch thick aluminum foil.

3/ Alternatively hot sauce may be packed loose in the menu bag.

3.2 Accessory packet assembly. One each of the components indicated in table II shall be inserted in an accessory packet as specified in 3.1.2. The maximum thickness of the filled and sealed accessory packet shall be 7/8 inch for accessory packet A and 1 inch for accessory packet B. When a preformed packet is used (see 3.1.2.1), a minimum of 1/8 inch wide heat seal shall be applied to effect closure of the packet. When a form-fill-seal packet is used (see 3.1.2.2), components shall be placed in the tray-shaped or flat sheet body and the cover applied by heat sealing. The sealed accessory packets shall not show any evidence of material degradation or delamination from packet fabrication, forming, or heat sealing. The average seal strength of the packet seals shall be not less than 3.5 pounds per inch of width and no individual specimen shall have a seal strength of less than 3.0 pounds per inch width when tested in accordance with 4.2.2.2.

3.3 Menu assembly. One each of the applicable components indicated in table III shall be inserted in the meal bag specified in 3.1.1. The components shall be arranged to produce a complete unit menu pack approximately 9 inches long by 5 inches wide by 2 inches thick. Excess air shall be expelled prior to closure sealing to allow packing as specified in 5.1 without damaging any of the meal bags or components therein. The open end of the bag shall be hermetically sealed with a heat seal not less than 1/16 inch wide. The closure seal shall have an average seal strength of not less than 9 pounds per linear inch and no individual specimen shall have a seal strength of less than 7 pounds (see 4.2.5.2).

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TABLE III. Menu contents

Menu No., Components therein	Menu No., Components therein
Menu #1	Menu #2
Pork with rice in barbecue sauce Applesauce Jelly Crackers Cocoa beverage powder Sugar-free beverage base Hot sauce Accessory packet, B Spoon Flameless heater	Corned beef hash Freeze-dehydrated fruit 1/ Jelly Crackers Oatmeal cookie bar Cocoa beverage powder Sugar-free beverage base Hot sauce Accessory packet, B Spoon Flameless heater
Menu #3	Menu #4
Chicken stew Wet pack fruit 2/ Peanut butter Crackers Candy 3/ Beverage base, powdered Hot sauce Accessory packet, A Spoon Flameless heater	Omelet with ham Potatoes au gratin Jelly Crackers Candy 3/ Oatmeal cookie bar Cocoa beverage powder Sugar-free beverage base Hot sauce Accessory packet, A Spoon Flameless heater
Menu #5	Menu #6
Spaghetti with meat and sauce Pound cake 4/ Cheese spread Crackers Candy 3/ Cocoa beverage powder Sugar-free beverage base Hot sauce Accessory packet, A Spoon Flameless heater	Frankfurters, beef Potato sticks Freeze-dehydrated fruit 1/ Peanut butter Crackers Candy 3/ Cocoa beverage powder Sugar-free beverage base Hot sauce Accessory packet, A Spoon Flameless heater

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TABLE III. Menu contents (cont'd)

Menu No., Components therein	Menu No., Components therein
Menu #7 Beef stew Pound cake 4/ Peanut butter Crackers Beverage base, powdered Hot sauce Accessory packet, A Spoon Flameless heater	Menu #8 Ham slice Potatoes au gratin Brownie, chocolate covered Cheese spread Crackers Cocoa beverage powder Beverage base, powdered Hot sauce Accessory packet, A Spoon Flameless heater
Menu #9 Pork chow mein Chow mein noodles Freeze-dehydrated fruit 1/ Peanut butter Crackers Cookie, chocolate covered Beverage base, powdered Hot sauce Accessory packet, A Spoon Flameless heater	Menu #10 Tuna with noodles Pound cake 4/ Jelly Crackers Candy 3/ Beverage base, powdered Hot sauce Accessory packet, A Spoon Flameless heater
Menu #11 Chicken and rice Cheese spread Crackers Candy 3/ Cookie, chocolate covered Beverage base, powdered Hot sauce Accessory packet, A Spoon Flameless heater	Menu #12 Escalloped potatoes with ham Applesauce Brownie, chocolate covered Cheese spread Crackers Cocoa beverage powder Sugar-free beverage base Hot sauce Accessory packet, A Spoon Flameless heater

1/ At least three of the four different kinds of freeze-dehydrated fruit (fruit mix, peaches, pears, strawberries) shall be distributed in each shipping container.

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2/ Type III, IV, or V of MIL-F-44067.

3/ Each shipping container shall have six menus with candy packed directly into the menu bag, outside of the accessory packet. The candy shall be procured so that three of the six menus contain packages of chocolate disks, two of the six menus contain heat stable chocolate bars, and one of the menus contains a package of fruit tablets.

4/ Types VII, VIII, IX, X, and XI shall be procured in equal quantities and distributed in as uniform a distribution as possible in each shipping container. However, not more than two meal bags in any shipping container shall contain the same type of cake.

3.4 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods, and the plant sanitation requirements of the appropriate Government inspection agency.

3.5 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.1.1 Material certification. Material listed below shall be accepted on the basis of a contractor's certification of conformance to the indicated requirements.

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Requirement	Requirement paragraph	Test procedure
Meal bag material:		
Identification	3.1.1	Laboratory evaluation
Thickness	3.1.1	Laboratory evaluation
Color	3.1.1	Visual evaluation
Accessory packet material:		
Identification	3.1.2.1 and 3.1.2.2	Laboratory evaluation
Polyethylene, linear low density polyethylene, polypropylene and polyester film thickness	3.1.2.1 and 3.1.2.2	As specified in L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378.
Aluminum foil thickness	3.1.2.1 and 3.1.2.2	As specified in QQ-A-1876
Color	3.1.2.1 and 3.1.2.2	Visual evaluation

4.2.1.2 Unfilled meal bag and preformed accessory packet certification. A certificate of compliance may be accepted as evidence that unfilled bags and packets conform to the requirements specified in 3.1.1 and 3.1.2.1, respectively. When deemed necessary by the inspection agency, testing of the unfilled bags and packets for seal strength of manufacturer's seals shall be as specified in 4.2.1.2.1 and 4.2.1.2.2.

4.2.1.2.1 Unfilled preformed accessory packet seal strength testing. The unfilled preformed packets shall be tested for seal strength in accordance with ASTM F 88 except that the specimen holding clamps shall be spaced 2 inches apart prior to testing and the testing speed shall be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. One-half or 1 inch wide test specimens shall be cut to a length suitable for proper mounting. The lot size shall be expressed in packets. The sample unit shall be one unfilled, preformed accessory packet. The inspection level shall be S-1 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 10.0. Three adjacent specimens 1/2 or 1 inch wide, shall be cut from each of the three sealed sides of each packet in the sample. The average seal strength of each seal shall be calculated by averaging the test results of the three test specimens. The results shall be reported to the nearest 0.1 pound per inch of width. Any test specimen failing to meet the seal strength requirements specified in 3.1.2.1 shall be scored as a major defect. Failure to meet the average seal strength requirement shall be cause for rejection of the lot.

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4.2.1.2.2 Unfilled meal bag seal strength testing. The unfilled bags shall be tested for seal strength in accordance with ASTM F 88 except that the specimen holding clamps shall be spaced 2 inches apart prior to testing and the testing speed may be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. One half or 1 inch wide test specimens shall be cut to a length suitable for proper mounting. Three adjacent specimens 1/2 or 1 inch wide, shall be cut from the sealed side of each bag in the sample. The results shall be reported to the nearest 0.1 pound per inch of width. The average seal strength shall be calculated by averaging the three test specimens cut from that seal. Any test specimen or average seal strength failing to meet the requirements of 3.1.1 shall be classified as a major defect and shall be cause for rejection of the lot. The lot size shall be expressed in bags. The inspection level shall be S-1.

4.2.2 Accessory packet inspection.

4.2.2.1 Accessory packet examination. The filled and sealed preformed packets or form-fill-seal packets, shall be examined for the defects listed in table IV. The lot size shall be expressed in packets. The sample unit shall be one filled and sealed packet. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for minor defects.

TABLE IV. Filled and sealed accessory packet defects

Category		Defect
Major	Minor	
101		Not clean 1/
	201	Effective heat seal closure width less than 1/16 inch at any point
	202	Thickness of packet greater than specified
	203	Tear, nick, notch, or serrations missing; or do not facilitate easy opening; or assembler's seal extends into or below tear, nick, or notch
	204	Dimensions not as specified
	205	Tear, hole, or open seal
	206	Evidence of delamination
	207	Labeling missing, incorrect, or illegible

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1/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause package damage (for example, glass, metal filings, etc.), or generally detracts from the clean appearance of the package. The following examples shall not be scored as defects for unclean:

a. Foreign matter which presents no health hazard or potential package damage and which can be readily removed by gently shaking the package or by gently brushing the package with a clean dry cloth.

b. Localized dried product which affects less than 1/8 of the total surface area of one pouch face, or an aggregate of scattered dried product which affects less than 1/4 of the total surface area of one pouch face.

4.2.2.2 Accessory packet closure seal testing. The filled and sealed packets shall be tested in accordance with ASTM F 88, except that the specimen holding clamps shall be spaced 2 inches apart prior to testing and the testing speed may be 10 or 12 inches per minute. The test specimen length shall be suitable for proper mounting. For preformed packets, three adjacent specimens, 1/2 or 1 inch wide, shall be cut from the closure seal of each packet in the sample. For the form-fill-seal packets, three adjacent specimens, 1/2 or 1 inch wide, shall be cut from each side and each end of each packet in the sample. The average seal strength of the closure seal for the preformed packet shall be calculated by averaging the test results of the three test specimens cut from that seal. For the form-fill-seal packet, the average seal strength of each side and end seal of the packet shall be calculated by averaging the test results of the three specimens cut from each side seal or end seal. The results shall be reported to the nearest 0.1 pound per inch of width. The lot size shall be expressed in packets. The sample unit shall be one sealed packet. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 10.0. Any test specimen failing to meet the requirements of 3.2 shall be classified as a major defect. Failure to meet the average seal strength requirements shall be cause for rejection of the lot.

4.2.2.3 Accessory packet and contents examination. The filled and sealed packets shall be examined for the defects listed in table V (this examination may be performed on the preformed packets after filling and prior to sealing). The lot size shall be expressed in packets. The sample unit shall be one packet open or sealed. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 4.0 for minor defects.

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TABLE V. Accessory packet and contents defects

Category		Defect
Major	Minor	
101		Missing or crushed candy
102		Foreign odor
103		Component not clean 1/
104		Plastic shrink film missing from around screw cap of hot sauce bottle or hot sauce bottle leaking
	201	Missing (except candy) or unserviceable component (for example, tear, hole, or open seam in coffee, cream substitute, sugar, salt, or hand cleaner package; or crushed gum)

1/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause package damage (for example, glass, metal filings, etc.), or generally detracts from the clean appearance of the package. The following examples shall not be scored as defects for unclean:

a. Foreign matter which presents no health hazard or potential package damage and which can be readily removed by gently shaking the package or by gently brushing the package with a clean dry cloth.

b. Localized dried product which affects less than 1/8 of the total surface area of one package face, or an aggregate of scattered dried product which affects less than 1/4 of the total surface area of one package face.

4.2.3 Cracker inspection. When crackers are vacuum sealed in the laminated bags by the unit packager, the unit packager shall be responsible for meeting the applicable quality assurance provisions of MIL-C-44112.

4.2.4 Shipping container inspection.

4.2.4.1 Shipping container examination. Examination of the filled and closed shipping containers shall be in accordance with the examination criteria of the appendix to PPP-B-636 except that the closure examination shall be in accordance with 4.2.4.2. In addition, the following defects shall be included in the table of examination:

Major - National stock number, item description, contract number, lot number or date of pack markings missing, incorrect, or illegible.

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- Metal fasteners used in closing the box.
- Outer flaps do not completely meet, leaving an opening greater than 3/4 inch between flap ends.

Minor - Other required markings missing, incorrect, or illegible.

4.2.4.2 Flap closure testing. The closure of the four outer flaps of the container shall be tested separately. A 90 degree angular bar with each leg approximately 5 inches long by 3 inches wide by 1/8 inch thick shall be used to test the flap closures. Insert one leg of the angular bar full length under the center of one outer flap. Insertion shall be made through the open slot between the outer flaps. Lift the container vertically by the other leg of the bar until the container is suspended. The complete upper surface of the inserted leg shall be in contact with the inner surface of the flap during the lifting and suspension of the container. Complete separation of the adhesive bond of one or more of the outer flaps, showing no evidence of fiber tear, shall be scored as a major defect. The lot size shall be expressed in shipping containers. The sample unit shall be one shipping container. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.2.4.3 Shipping container interior examination. The filled and sealed shipping containers shall be examined for the defects listed in table VI. The lot size shall be expressed in shipping containers. The sample unit shall be one filled and sealed shipping container. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0 for major defects and 10.0 for minor defects.

TABLE VI. Shipping container interior defects

Category		Defect
Major	Minor	
101		Less than 12 menus or not one of each
	201	Not packed or arranged as specified

4.2.4.4 Examination of adhered sleeve joint. A certificate of compliance may be accepted as evidence that the sleeve glue joint conforms to the requirements for weather-resistant boxes, glue closure examination, as specified in the appendix of PPP-B-636.

4.2.5 Meal bag inspection.

4.2.5.1 Meal bag examination. The filled and sealed meal bags shall be examined for the defects listed in table VII. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed meal bag.

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3.4.4.4.4 Types I and II, Classes 1A, 1B, 2, 3A, and 3B, all grades.

The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for minor defects. A minimum of 50 samples shall be examined for critical defects. The finding of any critical defect shall be cause for rejection of the lot. The sample meal bags shall be selected from shipping containers which have been filled, sealed, sleeved, and strapped. The inspection sample shall contain a proportionate amount of each of the 12 menus.

TABLE VII. Meal bag defects

Category			Defects
Critical	Major	Minor	
1			Tear, hole, or puncture through carton or open carton causing a hole in the pouch or obviously wet or stained carton due to leaking pouch 1/
2			Tear, hole, or puncture in cheese spread pouch
3			Swollen cheese spread pouch, or swollen pouch or carton of thermostabilized items
	101		Tear or hole in carton exposing pouch to potential damage 1/
	102		Menu component missing or incorrect assortment for menu package 2/
	103		Foreign odor
	104		Not clean - the meal bag or any of the outer packaging of its contents 3/
	105		Labeling missing, incorrect, or illegible
	106		Tear, hole, puncture, open seal or swollen pouch in peanut butter or jelly
	107		Tear, hole, puncture, or open seal in cocoa beverage powder bag, candy package, beverage base bag, cake bag, potato sticks bag, chow mein noodles bag, or ration heater bag
	108		Loss of vacuum - cracker bag, dehydrated fruit block, cookie, oatmeal cookie bar, or brownie bag 4/

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TABLE VII. Meal bag defects (cont'd)

Category			
Critical	Major	Minor	Defects
	109		Crushed or broken cracker; crushed dehydrated fruit block, cookie, oatmeal cookie bar, brownie, or accessory packet adversely affecting candy or hot sauce; crumbled, crushed, or broken ration heater pad 5/
	110		Broken spoon
	111		Chocolate toffee rolls or vanilla caramels not packed in aluminum foil barrier bag or B accessory packet
		201	Tear notch or slit missing or not as specified
		202	Tear, hole, open seal, or split in meal bag
		203	Thermostabilized carton flaps open or tear or hole in carton not exposing pouch to potential damage
		204	Tear, hole, puncture, or open seal in accessory packet bag

1/ Applies to cartoned items.

2/ Hot sauce shall be either loose in the menu bag or packed in accessory packet bag.

3/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause package damage (i.e. glass, metal filings, etc.), or generally detracts from the clean appearance of the package. The following examples shall not be scored as defects for unclean:

a. Foreign matter which presents no health hazard or potential package damage and which can be readily removed by gently shaking the package or by gently brushing the package with a clean dry cloth.

b. Localized dried product which affects less than 1/8 of the total surface area of one pouch face, or an aggregate of scattered dried product which affects less than 1/4 of the total surface area of one pouch face.

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c. Water spots.

d. Very thin film of grease, oil, or product residue which is discernible to touch, but is not readily discernible by visual examination.

4/ When vacuum retention cannot be determined visually by obvious cling of the bag walls to the contents, retention shall be verified by testing as specified in the applicable component specification.

5/ Examination for crushed product shall be performed on closed component bags. A bag containing crushed, or broken crackers shall be indicated by any bag that contains a cracker less than 3-3/4 inches in length or width (indicating crushing of the length or width); that has any depression (other than docker holes or serrations) when the bag is passed through the fingers while exerting sufficient finger grip pressure to feel the serrated areas (indicating crushing in thickness); that has movement of the cracker particles felt through the bag (indicating broken or crushed crackers); or that has any permanent wrinkling of the bag over the cracker surface (indicating crushed or broken crackers and bag contraction resulting from the bag vacuum). A crushed dehydrated fruit block, cookie, oatmeal cookie bar, or brownie is one in which 1/8 or more of the volume of the item has been reduced to crumbs or powder. A crumbled, crushed, or broken heater pad is one in which the paperboard cover is broken open and spilling its contents or the paperboard cover is misshapen as the result of having been crushed, bent, or otherwise mishandled during the ration assembly operation. The presence of loose metal fragments and/or powders that are randomly distributed throughout the interior of the bag as the apparent result of static electric forces shall not be considered evidence of crumbled, crushed, or broken heaters.

4.2.5.2 Meal bag closure seal strength testing. The filled and sealed meal bags shall be tested for closure seal strength in accordance with ASTM F 88 except that the specimen holding clamps shall be spaced 2 inches apart prior to testing and the testing speed may be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. One half or 1 inch wide test specimens shall be cut to a length suitable for proper mounting. Three adjacent specimens, 1/2 or 1 inch wide, shall be cut from the closure seal of each bag in the sample. The results shall be reported to the nearest 0.1 pound per inch of width for each specimen and for the average of the three specimens from each closure seal. Any test specimen or average seal strength failing to meet the requirements of 3.3 shall be classified as a major defect and shall be cause for rejection of the lot. The lot size shall be expressed in bags. The inspection level shall be S-1.

4.2.6 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

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5. PACKAGING

5.1 Packing. Twelve meals, one of each menu of table III shall be packed in a fiberboard box with box liner and sleeve constructed and closed in accordance with style RSC-SL, grade V2s of PPP-B-636, except metal fasteners shall not be used in the closure of the flaps, the gap between the outer flaps shall be not more than 3/4 inch wide, and adequacy of the closure shall be determined by testing as specified in 4.2.4.2. Inner flaps of the box may be closed with two strips of hot melt adhesive per flap, provided the closure meets the test requirement of 4.2.4.2. As an alternative to body joint metal fasteners, adhesive conforming to MMM-A-250, applied in accordance with "adhesive joint" requirement of PPP-B-636, may be used to secure the manufacturer's joint of the V2s box. As an alternative to sleeve joint metal fasteners, adhesive conforming to MMM-A-250 may be used for securing the joint, except the overlap shall be a minimum of 3 inches wide. The adhesive joint shall be examined in accordance with 4.2.4.4. The adhesive shall cover a minimum of 90 percent of the sleeve overlap area. Meal bags shall be packed flat or on edge, and arranged as specified in figure 1. Boxes shall be reinforced in accordance with the appendix of PPP-B-636, except pressure-sensitive adhesive filament reinforced tape is prohibited and boxes shall be reinforced with two girthwise straps positioned to divide the box into units of equal length. The inside dimensions of the fiberboard box shall be 15-9/16 inches in length, 9-1/8 inches in width, and 9-5/8 inches in depth.

5.2 Unit loads. The Meal, Ready-To-Eat, Individual, packed as specified in 5.1, shall be arranged in unit loads in accordance with type I, class B of MIL-L-35078.

5.3 Labeling and marking.

5.3.1 Shipping containers. Shipping containers for complete MREs and those for shipping components to the ration assembler shall be marked in accordance with MIL-STD-129 and as specified in the contract with the following exceptions:

a. Identification markings normally placed on an end of the shipping container shall be placed on the corresponding end of the sleeve. The marking on the sleeve shall read from top to bottom, left to right, when the shipping container is rotated from its upright position onto its side for palletization. The side panel of the sleeve immediately to the right of the marked end of the sleeve shall bear the following markings:

- Contract data and other required markings
- Date of pack
- Lot number
- NOTE: WATER ACTIVATED Flameless Ration Heater, NSN 8970-01-321-9153, supplied in each MRE menu bag.

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b. One side panel of the shipping container shall be marked "MEAL, READY-TO-EAT, INDIVIDUAL" in letters not less than 1-1/4 inches high. Underneath the ration nomenclature, in letters not less than 1 inch high, the shipping container shall be marked "DO NOT ROUGH HANDLE WHEN FROZEN (0°F or below)".

c. The bottom end of the sleeve as well as the bottom of the box shall convey the box maker's compliance markings as required.

5.3.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078 and as specified in the contract. In addition, each unit load shall be provided with a Material Safety Data Sheet (MSDS), in accordance with MIL-R-44398. The MSDS shall be placed inside a clear plastic sleeve and securely attached to one side of the unit load with tape or pressure sensitive adhesive. A copy of the MSDS must be included with the shipping papers and a copy must also be placed in the vehicle manifest.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When chocolate covered brownies and cookies are supplied in bulk packs and flexible unit packing is to be done by the ration assembler (see table I).
d. When crackers are supplied in bulk packs and flexible unit packing is to be done by the unit packager (see table I).

6.2 Subject term (key word) listing.

Combat field feeding
Operational rations

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

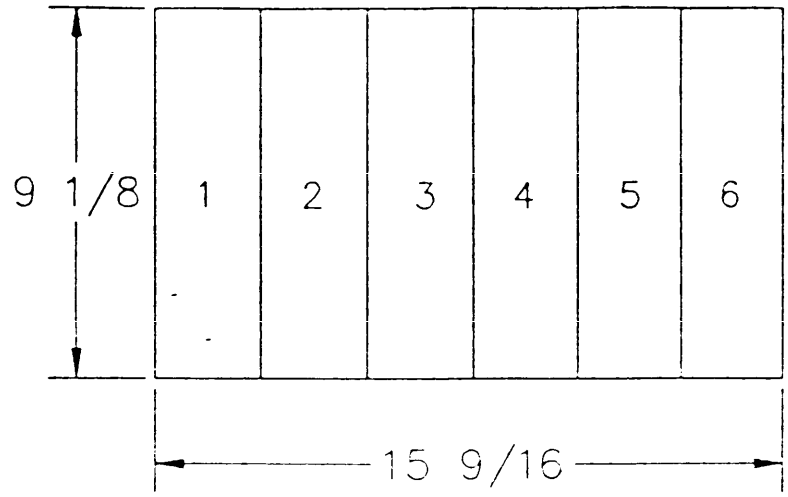
Army - MD, QM
Navy - MC
DP - SS

Preparing activity:

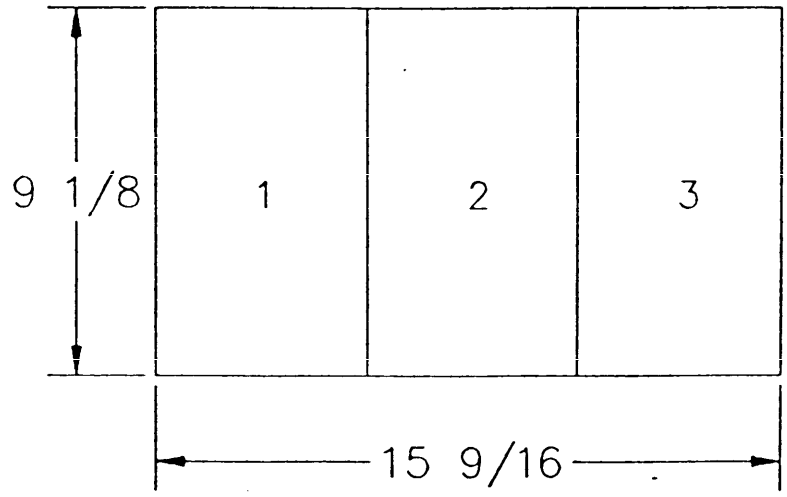
Army - GL

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MEAL BAGS ON EDGE
2 LAYERS



MEAL BAGS ON FLAT
4 LAYERS

FIGURE 1. ACCEPTED MEAL BAG PACKING CONFIGURATIONS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-44074E	1992 December 31

3. DOCUMENT TITLE

MEAL, READY-TO-EAT, INDIVIDUAL, ASSEMBLY OF

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4501	256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WTP Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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